Axie Economy Analysis

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1 Introduction

The Axie Infinity economy had it's SLP tied to real money and that's why its crash had consequences. What we need to consider is that even if it had not been tied to real money, it's effects would have arguably been atleast comparable to if it had been tied to money. Hyperinflation would have affected new players, making the game an unattractive play regardless.

Before we get into why it happened, we need to get into what happened. Axie had a play to earn model through which by playing you earn nfts which have some value in a player determined market. The main exchange token to earn in Axie was SLP(whose expansion is irrelevant to the goals of this analysis) and the problem with the economy is SLP lost over 99% of it's value which heavily discouraged trading and led to very low retention rates for the game.

2 Currency supply injection and extraction

The main resource that we are going to be concerned with is going to be the SLP and it's relation to the US dollar. Since the value of the SLP was directly tied to the USD, it is reasonable to assume that it functions like an actual currency and not just as an experiment with which users interact for entertainment.

Estimates for the injection rate per person into the economy is around 100-200.We can estimate the amount of currency each player earns. The currency sink for SLP was breeding of axies. Breeding took a certain number of SLP

Breed Count	Breed Number	SLP Cost Per Parent
(0/7)	1	900
(1/7)	2	1350
(2/7)	3	2250
(3/7)	4	3600
(4/7)	5	5850
(5/7)	6	9450
(6/7)	7	15300

Figure 1: Breeding chart

depending upon the generation of the Axie. We can assume that breeding axies took the median value for the number of times bred and hence the median SLP. This method has it's inaccuracies but as we are going to see, for the purposes of this analysis, the compounded effects eventually cancel out.

3 Mathematical modeling

Before we begin the modeling, we first need to make some reasonable assumptions.
1. Player growth starts off slow, increases and then reduces again. What this means mathematically is that the growth function resembles is a bell curve.
2. The propensity to breed axies is directly proportional to the the player

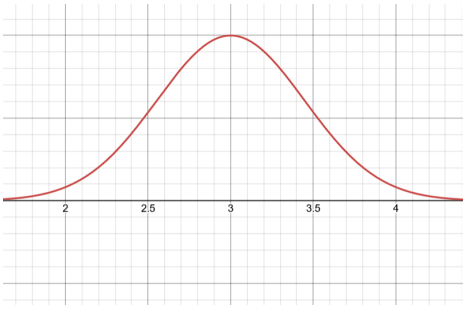


Figure 2: Bell Curve

growth, which is when there is actual demand in the resale market for axies. This is a reasonable estimation because it is presumed older players already have the axies with the characteristics new players are looking for. The effects at the more expensive end of the characteristics is going to be delayed but they will also experience the growth in the market and hence growth in demand.
Using these, we can begin to model the circumstances that led to the hyperinflation of SLP.
1.The thing that we have to model first is the player count for the game. This is simply a matter of integration of the player growth statistic. One things to consider is that the median of our bell curve is positive and reasonably away from from (0,0) such that in comparison the y value of the bell curve when x = 0 is approximately 0.

$$y = e^{-\frac{(x-a)^2}{b}}$$

The total player function turns out to be

$$T = \int e^{-\frac{(x-a)^2}{b}}$$

which approximately looks like a sigmoid function

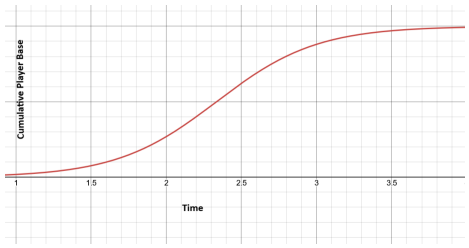


Figure 3: Sigmoid function

2. What the player earns everyday can be modeled as

$$E = 25 + r$$

Where r is a random point taken from a gaussian distribution with r being a function of luck, skill, time ranging from a minimum of 0 with the median centered around 150 with a standard deviation of 25.
The addition to the circulating supply can be modeled as

$$M = (\int e^{-\frac{(x-a)^2}{b}}) \times (25 + r)$$

3. The propensity to breed can be

$$P = \frac{(e^{-\frac{(x-a)^2}{b}})}{(\int e^{-\frac{(x-a)^2}{b}}) - (e^{-\frac{(x-a)^2}{b}})}$$

which is the ratio of new population to the existing population.
The daily sink into breeding axies is

$$S = k\frac{(e^{-\frac{(x-a)^2}{b}})}{(\int e^{-\frac{(x-a)^2}{b}}) - (e^{-\frac{(x-a)^2}{b}})} * 7200$$

where k is the proportionality constant assumed as 1 and 7200 is the cost to breed two median axies
4. The increase or decrease to the money supply is

$$I = ((\int e^{-\frac{(x-a)^2}{b}}) \times (25 + r)) - (k\frac{(e^{-\frac{(x-a)^2}{b}})}{(\int e^{-\frac{(x-a)^2}{b}}) - (e^{-\frac{(x-a)^2}{b}}))} * 7200)$$

5. The new total money supply is

$$N = Q + ((\int e^{-\frac{(x-a)^2}{b}}) \times (25 + r)) - (k\frac{(e^{-\frac{(x-a)^2}{b}})}{(\int e^{-\frac{(x-a)^2}{b}}) - (e^{-\frac{(x-a)^2}{b}}))} * 7200)$$

Where Q is initial supply

4 Code

The python code used for this project is

```
import numpy as np
import csv
rng = np.random.default_rng()

failure_details = []

i = 0

def MCS(sim = 1000, days = 200):

     failure = 0



     for i in range(sim):
          users = 10000
          supply = 1000000
          a= rng.uniform(10, 100)
          b= max(1, rng.uniform(100, 1600))



          baseline_burn = rng.uniform(20, 75)
          for day in range(days):
             player_growth = 500*np.exp(-((day - a)**2) / b)
             users = users + player_growth
             daily_mint_user = max(25, rng.normal(loc = 150, scale = 25))
             daily_mint = daily_mint_user * users
             propensity = player_growth / ( users- player_growth )
             daily_sink = (propensity * 7200 + baseline_burn)*users
             deltaM = daily_mint - daily_sink
             supply = supply + deltaM
             if supply/users >10000:
                 failure += 1
                 deets = [ int(supply), int(users), int(baseline_burn), int(a), int(b) ]
                 failure_details.append(deets)

                 print("Failure on simulation ", i, " on day ", day, )


                 break
     print(failure)
```

```
def csv_exp():
    with open('failure_details.csv', 'w', newline='') as f:
        writer = csv.writer(f)
        writer.writerow(['Supply', 'Users', 'Baseline Burn', 'a', 'b'])
        writer.writerows(failure_details)
```

```
MCS()
csv_exp()
```

5 Statistical Results

Here, we are going to use python to run Monte Carlo Simulations and map the possible outcomes. Let us establish a failure condition such that when

$$\frac{Q}{T} > 10000$$

Our initial conditions are going to be
Initial users = 10,000 and Initial money supply = 1,000,000.
Baseline currency burn per user can be anywhere from 20 to 75
Maximum player growth on any day cannot be greater than 500
a and b are uniformly chosen to simulate different peaks and different growth periods

There were a total 999 simulations out of which 658 were failures.

1. The plot of b versus the number of failures looks like figure 4

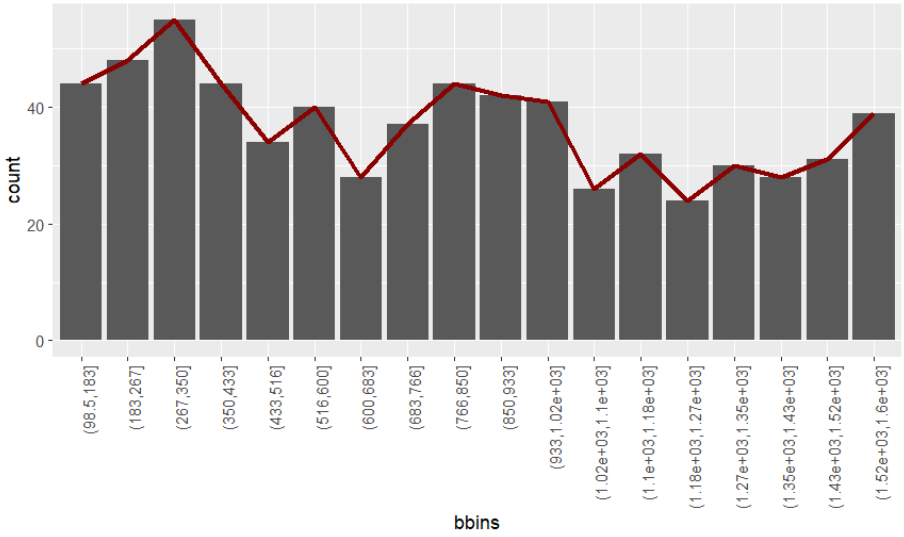


Figure 4: b with number of failures

There seems to be a weak correlation with increasing values of b and failure rate, but nothing conclusive

2.The plot of a versus the number of failures looks like figure 5
Here, there seems to be a much stronger correlation between failure rate and lower values of a. This is possibly due to the fact that a controls the peak of the gaussian and that those with higher values of a might also lead to failure, just at a day past 200.

3. The plot of baseline versus the number of failures looks like figure 6
This plot follows the logical and expected pattern that failures are less likely if the baseline burn value is higher, this values being too high might lead to failure in the other direction.

4.The plot of a and b versus the number of failures looks like figure 7
There is a weak correlation in the figure to indicate lower values of both a and b leading to a higher number of failures.

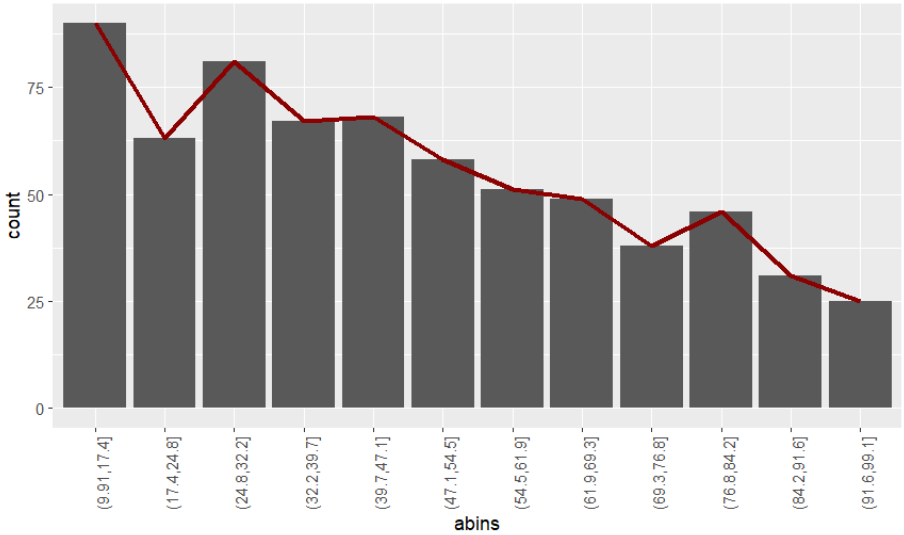


Figure 5: a with number of failures

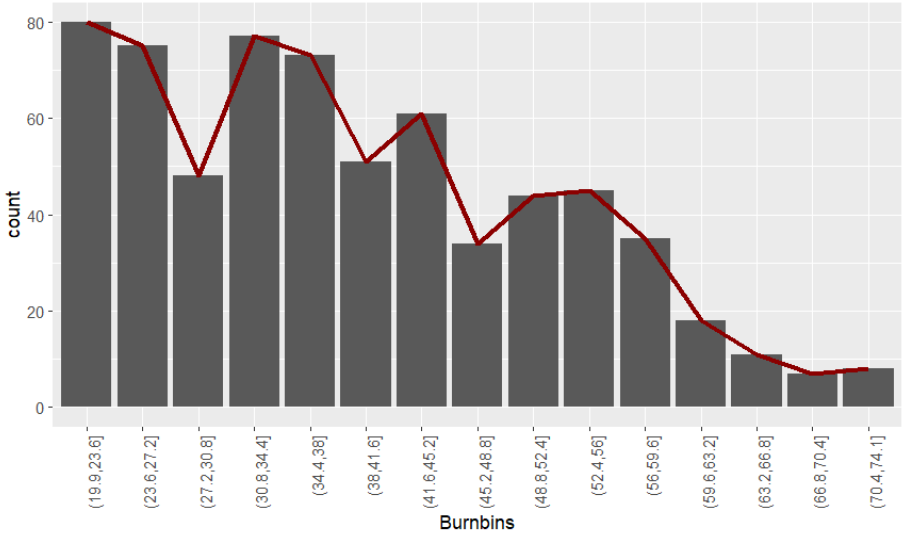


Figure 6: Baseline burn with number of failures

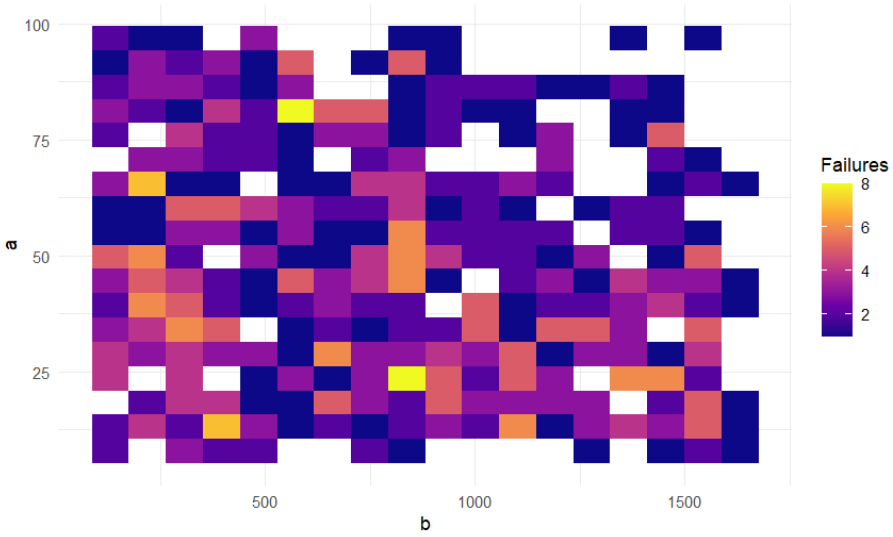


Figure 7: a and b

6 Conclusion

These results indicate to us that there do exist some correlations between the variables given above and the phenomenon of hyperinflation. The question that remains is can this hyperinflation be avoided ?. My answer is yes but not through tweaking variables. My solution would be to implement a sink that scales with the total number of users rather than the growth rate of users. There is definitely more that can be said about these circumstances, but this paper limits itself and hopes to set a precedent for future analysis.